Supplemental Material for "Interband Pairing Signatures in the Superfluid Response of Magic-Angle Twisted Bilayer Graphene"

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TABLE I. Default parameters used for the mechanism scans.

quantity	value
twist angle	$\theta = 1.05^\circ$
Dirac velocity	$v_F = 2.135$ eV Å
AA tunneling	$w_0 = 87.2$ meV
AB tunneling	$w_1 = 109.0$ meV
reciprocal cutoff	$N_{\text{shell}} = 3$
retained bands	$n_{\text{keep}} = 6$
momentum grid	$nk = 7$ for dense maps, $nk = 9, 11, 13$ for key-point checks
gap scale	$\Delta_0 = 1$ meV
velocity finite difference	$dk = 10^{-6}$ Å^{-1}

I. SCOPE OF THE SUPPLEMENTAL MATERIAL

This Supplemental Material documents the executable numerical workflow behind the PRB manuscript. It is organized as a reproducibility and self-audit record: the Bistritzer-MacDonald (BM) model conventions [1], the orbital-projected pairing construction, the finite-band response formula, the figure and table provenance, and the historical PRB benchmark reconstruction are recorded separately.

The main text makes a conservative mechanism claim: an orbital-projected interband-pairing direction produces a reproducible, normalization-conditioned change in the normalized finite-band superfluid response. The present Supplemental Material does not promote the raw response units to final experimental stiffness values. This choice is motivated by the distinction between robust quantum-geometric mechanism diagnostics [2–5] and direct quantitative comparison with recent stiffness experiments [6, 7].

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II. BM MODEL AND NUMERICAL PARAMETERS

The normal-state calculation uses the single-valley BM continuum Hamiltonian with orbital basis

$$(tA, tB, bA, bB), \tag{1}$$

where t, b label layers and A, B label sublattices. Unless explicitly stated otherwise, the production mechanism scans use the parameters in Table I.

The moire unit-cell area and reciprocal scale used in the unit audit are

$$A_M = 1.5606 \times 10^4 \ \text{Å}^2, \qquad G_M = 0.054047 \ \text{Å}^{-1}. \tag{2}$$

The raw response conversion used in the audit is

$$D[\text{eV Å}^2] = \frac{D_{\text{raw}}[\text{meV Å}^2]}{1000}, \tag{3}$$

with all remaining spin, valley, or convention factors kept explicit rather than absorbed into a hidden prefactor.

A. Reporting Convention for Units and Filling

Table II states the reporting convention used in the present manuscript. The main figures use normalized response ratios, while absolute eV Å^2 values are restricted to the self-audit and historical benchmark reconstruction.

TABLE II. Reporting convention for response and filling quantities.

quantity	present use	excluded interpretation
raw response	internal finite-band diagnostic	physical stiffness
$D_{\text{iso}}(\eta)/D_{\text{iso}}(0) - 1$	main mechanism observable	absolute stiffness
eV Å^2 per flavor	benchmark audit only	experimental comparison
ν_{proxy}	retained-band occupancy label	calibrated carrier density

The retained-band filling proxy is computed as

$$\nu_{\text{proxy}} = g \left[\left\langle N_{\text{keep}}^{-1} \sum_n \Theta(\mu - \varepsilon_{n\mathbf{k}}) \right\rangle_{\mathbf{k}} - \frac{1}{2} \right], \tag{4}$$

with $g = 4$ used as a spin-valley counting factor in the proxy. For the $n_{\text{keep}} = 6$ scans this is not identical to a calibrated experimental filling factor; it is used only to order the chemical-potential scans.

III. ORBITAL-PROJECTED PAIRING CONSTRUCTION

The pairing direction is specified first in the layer-sublattice orbital basis,

$$\Delta_{\text{orb}}(\eta) = \Delta_0 \mathcal{N}_\eta (M_0 + \eta M_1), \tag{5}$$

with

$$M_0 = \tau_0 \sigma_0, \qquad M_1 = \tau_x \sigma_x. \tag{6}$$

The two normalization controls used in the manuscript are

$$\text{fixed Frobenius:} \qquad \|M_0 + \eta M_1\|_F = \|M_0\|_F, \tag{7}$$

$$\text{fixed } \Delta_0 : \qquad \mathcal{N}_\eta = 1. \tag{8}$$

The band-basis pairing matrix is

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k}) \Delta_{\text{orb}} U_-^*(-\mathbf{k}). \tag{9}$$

The working valley convention in the present diagnostic is the time-reversal sewn proxy

$$U_-(-\mathbf{k}) = U_+^*(\mathbf{k}). \tag{10}$$

The production projection is therefore

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k}) \Delta_{\text{orb}} U_+(\mathbf{k}). \tag{11}$$

This convention is useful for isolating an orbital-defined interband-pairing signature. It also avoids a known gauge artifact: independently diagonalized valley-minus eigenvectors are not automatically in the same time-reversal sewing convention as the valley-plus eigenvectors. Table III summarizes the raw-basis diagnostic.

This is still a diagnostic convention rather than a full independent two-valley implementation. Valley-asymmetric perturbations, strain, or valley-dependent interactions would require revisiting this sewing choice.

The projected interband-pairing weight is

$$W_{\text{inter}} = \frac{\sum_{n \neq m} |\Delta_{nm}|^2}{\sum_{n,m} |\Delta_{nm}|^2}. \tag{12}$$

For the dense $nk = 7$, $n_{\text{keep}} = 6$ scan, $W_{\text{inter}} \simeq 0.108$ at $\eta = 1$.

TABLE III. Raw valley-minus sewing diagnostic. The independently diagonalized valley-minus subspace at $-\mathbf{k}$ is compared with the target $U_+^*(\mathbf{k})$ on an $nk = 3$ grid. Small singular values and large alignment errors indicate that the raw valley-minus eigenvectors should not be used directly to define W_{inter}.

n_{keep}	alignment error	mean minimum singular value	minimum singular value
2	1.3895	0.0267	0.0218
4	1.3782	0.0201	0.0043
6	1.3760	0.0095	0.0016

IV. FINITE-BAND RESPONSE AND CHANNEL SPLIT

For a retained band subspace, the finite-band BdG Hamiltonian is

$$H_{\mathrm{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) - \mu & \Delta_{\mathrm{band}}(\mathbf{k}) \\ \Delta_{\mathrm{band}}^\dagger(\mathbf{k}) & -[\varepsilon(\mathbf{k}) - \mu] \end{pmatrix}. \tag{13}$$

The zero-temperature static current-current response is evaluated as

$$\Lambda_{\alpha\beta} = \sum_{a \neq b} \frac{\langle a|J_\alpha|b\rangle \langle b|J_\beta|a\rangle (f_a - f_b)}{E_b - E_a}. \tag{14}$$

The band-basis velocity is split into diagonal and off-diagonal parts,

$$v_\alpha^{\mathrm{band}} = v_\alpha^{\mathrm{intra}} + v_\alpha^{\mathrm{inter}}. \tag{15}$$

This produces conventional, geometric, and cross-channel response diagnostics. The primary normalized response map uses

$$\delta D_{\mathrm{iso}}(\mu, \eta) = \frac{D_{\mathrm{iso}}(\mu, \eta)}{D_{\mathrm{iso}}(\mu, 0)} - 1, \qquad D_{\mathrm{iso}} = \frac{D_{xx} + D_{yy}}{2}. \tag{16}$$

V. FIGURE AND TABLE PROVENANCE

Table IV summarizes the current figure and table provenance. The full machine-readable record is maintained in `notes/figure_table_provenance_2026-06-09.md`.

The dense response data are reproduced by

TABLE IV. Manuscript artifact provenance. CSV values used in the manuscript tables are checked by `scripts/verify_manuscript_tables.py`.

artifact	data source	script	check
Frobenius heatmap	dense scan CSV	heatmap plotter	figure file exists
fixed-Δ_0 heatmap	dense scan CSV	heatmap plotter	figure file exists
dense response table	eta-summary CSV	eta summarizer	table verifier
$nk = 9$ validation table	key-point summary CSV	eta summarizer	table verifier
PRB audit table	reconstruction CSVs	audit scripts	table verifier

```
python3 scripts/run_mu_response_scan.py \
  --nk 7 --n-shell 3 --n-keep 6 \
  --mus -5 -4 -3 -2 -1 0 1 2 3 4 5 \
  --etas 0 0.25 0.5 0.75 1 \
  --output data/processed/mu_eta_response_scan_nk7_nkeep6.csv
```

and summarized by

```
python3 scripts/summarize_eta_effect.py \
  data/processed/mu_eta_response_scan_nk7_nkeep6.csv \
  --output data/processed/mu_eta_response_scan_nk7_nkeep6_eta1_summary.csv
```

The table verifier is

```
python3 scripts/verify_manuscript_tables.py
```

VI. KEY-POINT MOMENTUM-GRID CONVERGENCE

The dense maps in the main text use $nk = 7$, while selected key points were checked at $nk = 9$, $nk = 11$, and $nk = 13$. The higher-grid scans use

```
python3 scripts/run_mu_response_scan.py \
  --nk 11 --n-shell 3 --n-keep 6 \
  --mus -4 0 2 4 --etas 0 0.5 1 \
  --output data/processed/mu_response_scan_nk11_nkeep6_keypoints.csv
```

```
python3 scripts/run_mu_response_scan.py \
  --nk 13 --n-shell 3 --n-keep 6 \
  --mus -4 0 2 4 --etas 0 0.5 1 \
  --output data/processed/mu_response_scan_nk13_nkeep6_keypoints.csv
```

and is summarized with `scripts/summarize_eta_effect.py`. The convergence figure is generated with

```
python3 scripts/plot_nk_convergence_keypoints.py
```

Figure 1 and Table V compare the normalized $\eta = 1$ response relative to $\eta = 0$. The fixed-Frobenius response remains positive at all four key chemical potentials. The fixed-Δ_0 response remains small and sign-sensitive, consistent with the main-text interpretation that the interband-pairing signal is normalization-conditioned.

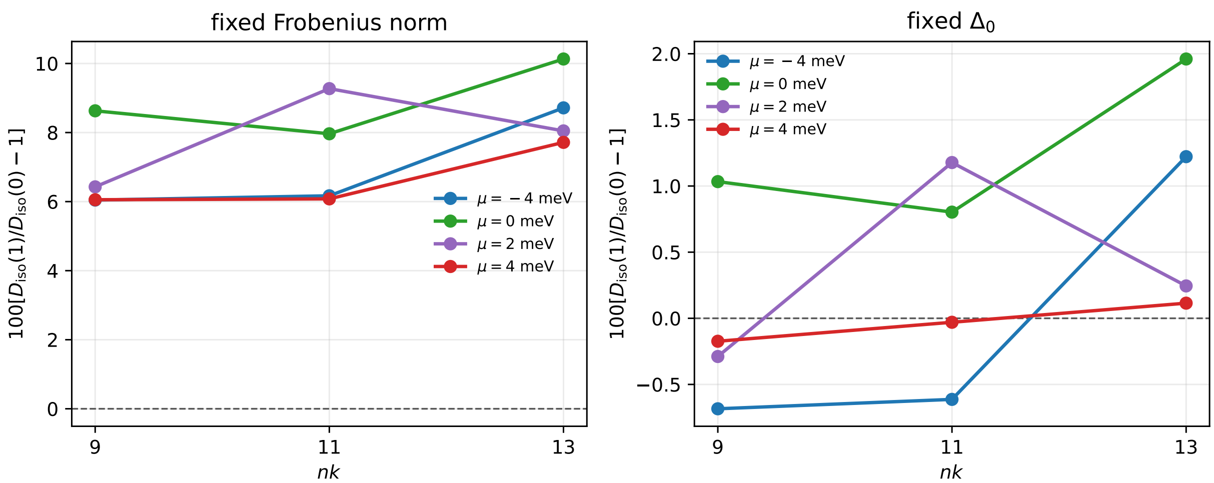

FIG. 1. Selected-grid convergence of the normalized $\eta = 1$ response. The left panel shows the fixed-Frobenius comparison used for the main mechanism claim; the right panel shows the fixed-Δ_0 control. The plotted values are drawn from the same summary data tabulated in Table V.

TABLE V. Key-point momentum-grid check for $D_{\rm iso}(\eta = 1)/D_{\rm iso}(\eta = 0) - 1$.

mode, μ (meV)	$nk = 9$	$nk = 11$	$nk = 13$	$13 - 9$
fixed Δ_0, -4	-0.0068	-0.0061	+0.0122	+0.0191
fixed Δ_0, 0	+0.0103	+0.0080	+0.0196	+0.0093
fixed Δ_0, 2	-0.0029	+0.0118	+0.0024	+0.0053
fixed Δ_0, 4	-0.0017	-0.0003	+0.0011	+0.0029
fixed Frobenius, -4	+0.0604	+0.0617	+0.0872	+0.0267
fixed Frobenius, 0	+0.0863	+0.0796	+0.1013	+0.0150
fixed Frobenius, 2	+0.0643	+0.0927	+0.0805	+0.0162
fixed Frobenius, 4	+0.0605	+0.0608	+0.0772	+0.0166

Across the three key-point grids the fixed-Frobenius response remains positive, with the largest $nk = 13 - nk = 9$ shift about 2.67 percentage points. Therefore the current manuscript claims sign and order-of-magnitude stability of the normalized mechanism signal, not a strict continuum-limit extrapolation of the tabulated values.

VII. HISTORICAL PRB BENCHMARK RECONSTRUCTION

The production interband-pairing scan and the historical PRB table reconstruction are kept separate. The best simple unified reconstruction route for the old uniform-s benchmark is

$$D_{\mathrm{conv}} = 2D_{\mathrm{intra},\tau_z}, \qquad D_{\mathrm{geom}} = D_{\mathrm{inter},\tau_z}. \tag{17}$$

For $n_{\mathrm{keep}} = 6$, this gives $D_{\mathrm{total}} = 126.66$ eV Å^2 versus the historical target 129.30 eV Å^2, and $D_{\mathrm{geom}} = 75.22$ eV Å^2 versus 75.30 eV Å^2.

The old $n_{\mathrm{keep}} = 2$ endpoint is more convention-sensitive. The closest audited reconstruction combines a Gamma-centered mesh, central-pair band selection, an all-τ_z current convention, a factor of two in the intraband term, and a full curvature-like conventional correction. This gives

$$D_{\mathrm{total}} = 66.18 \text{ eV Å}^2 \quad \text{versus} \quad 67.50 \text{ eV Å}^2. \tag{18}$$

Because this full-curvature correction does not work uniformly at $n_{\mathrm{keep}} = 6$, the reconstruction is treated as a historical audit rather than a production convention.

VIII. CURRENT LIMITATIONS AND NEXT VALIDATION GATES

The current submission-critical gaps are:

1. a systematic continuum-grid extrapolation beyond the present selected $nk = 9$, $nk = 11$, and $nk = 13$ key-point checks;

2. a fully independent two-valley implementation before studying valley-asymmetric perturbations;

3. a final absolute stiffness convention before any direct experimental comparison;

4. a calibrated filling variable replacing the retained-band proxy ν_{proxy}.

These limitations do not invalidate the normalized-response mechanism claim, but they prevent the current draft from being treated as a final PRB submission without further convergence and convention work.

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DECLARATIONS

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